

# PROJECT 1: Build a Home Lab Network

## Description

In this project, I created a virtualized network environment entirely on a personal computer. Using free software, I was able to simulate multiple computers, routers, and switches — and configured the networks to be able to communicate with each other just like a real corporate network. The objective of the lab will be to create a working multi-segment network with proper IP addressing, routing between subnets, and basic security controls.

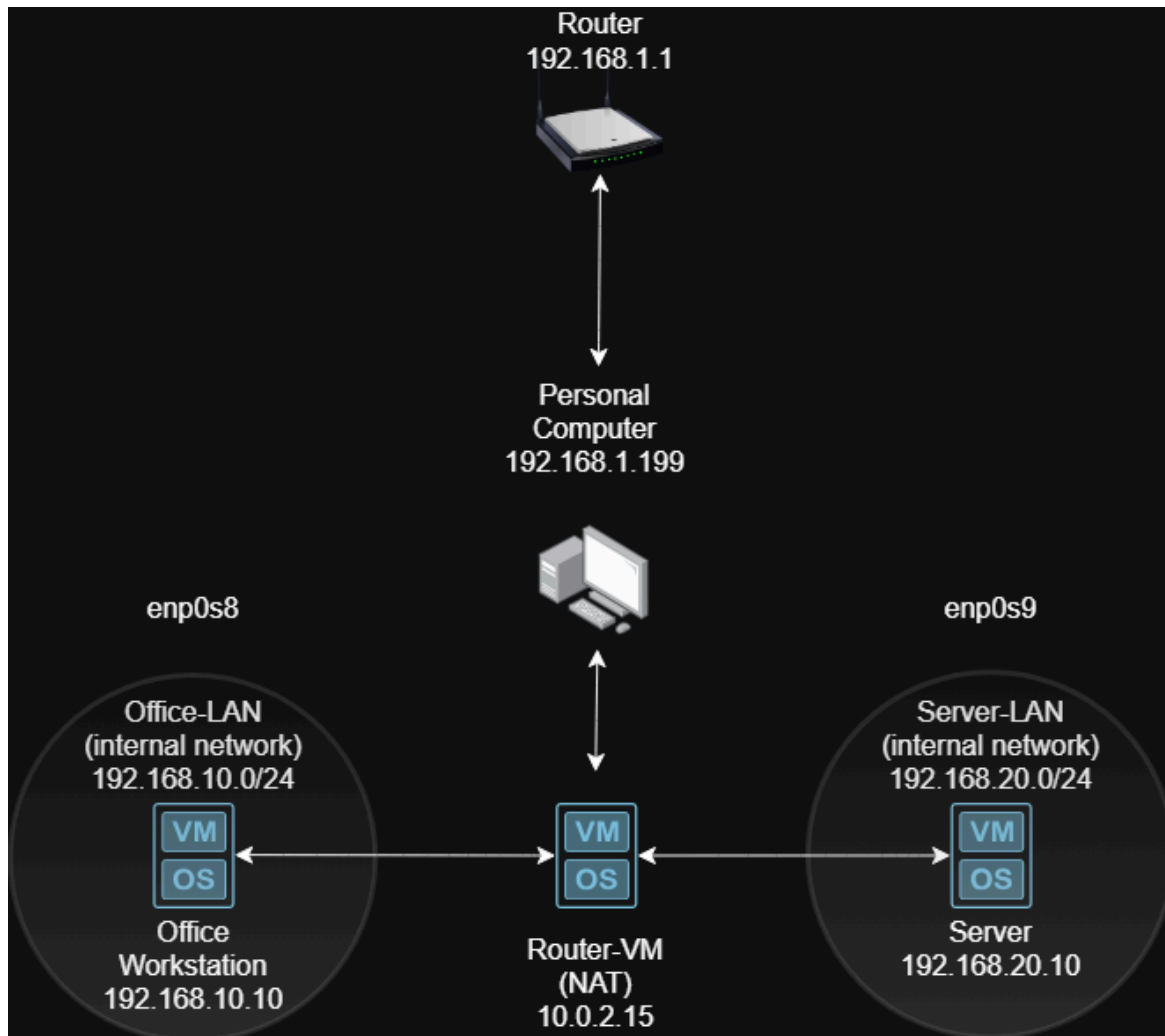
|  Core Concepts: Networking Fundamentals Used in This Project |
|-----------------------------------------------------------------------------------------------------------------------------------------------|
| • Virtualization                                                                                                                              |
| • IP Addressing (IPv4)                                                                                                                        |
| • Subnetting                                                                                                                                  |
| • CIDR Notation                                                                                                                               |
| • Default Gateway                                                                                                                             |
| • NAT (Network Address Translation)                                                                                                           |
| • DHCP                                                                                                                                        |
| • Network Segmentation                                                                                                                        |

## Tools Used

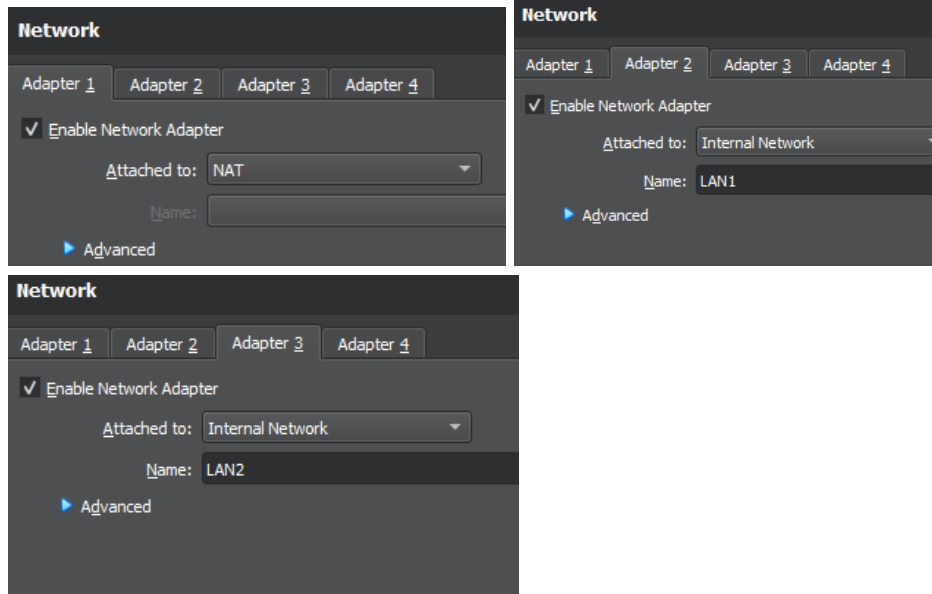
| Tool / Software         | Purpose                                   | Where to Get It                                                                                   |
|-------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------|
| VirtualBox              | Free hypervisor to run virtual machines   | <a href="https://www.virtualbox.org">virtualbox.org</a> (free download)                           |
| Ubuntu Server 22.04 ISO | Free Linux OS for your virtual machines   | <a href="https://ubuntu.com/download/server">ubuntu.com/download/server</a>                       |
| Windows 10/11 ISO       | Optional — for testing Windows clients    | <a href="https://microsoft.com/en-us/software-download">microsoft.com/en-us/software-download</a> |
| draw.io (diagrams.net)  | Free online tool to draw network diagrams | <a href="https://diagrams.net">diagrams.net</a>                                                   |

## Project Layout:

1. Using VirtualBox (type 2 hypervisor), installed Ubuntu 22.04 and created 3 different VMs, one for the router, one for the Office Workstation, and one for the Server
2. Created a network diagram to lay out the network architecture for all three hosts:



- I then configured three network interfaces, internet (NAT), and the two internal subnets on the Router-VM:



- Identified each network interface and edited the Netplan configuration on the Router-VM to enable IP forwarding between the two specified subnets: 192.168.10.1 and 192.168.20.0/24

```
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:80:d4:be brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86273sec preferred_lft 86273sec
    inet6 fe80::7584:d8e6:dd0f:d282/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:58:9e:74 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::75af:c664:bf0c:7708/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d7:31:b9 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::dc05:224f:aef9:8da4/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
GNU nano 7.2 /etc/netplan/00-installer-config.yaml
network:
  version: 2
  ethernets:
    enp0s8:
      addresses: [192.168.10.1/24]
    enp0s9:
      addresses: [192.168.20.1/24]
```

```
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$ sudo netplan apply
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$ echo 'net.ipv4.ip_forward=1' | sudo tee -a /etc/sysctl.
conf && sudo sysctl -p
net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
net.ipv4.ip_forward = 1
```

5. Manually configured network adapters for Office and Server VM for LAN 1 and LAN 2 as specified in our network diagram. IP addresses were statically assigned within the specified subnet. Ping tests were performed between Office (client) to the Server, office (client) to router, and server to router:

Office VM (192.168.10.10):

Ping Successful with Router-VM (10.0.2.15)

```
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=1.43 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.496 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.541 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.513 ms
64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=0.522 ms
64 bytes from 10.0.2.15: icmp_seq=6 ttl=64 time=0.631 ms
64 bytes from 10.0.2.15: icmp_seq=7 ttl=64 time=0.619 ms
^C
--- 10.0.2.15 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6082ms
rtt min/avg/max/mdev = 0.496/0.679/1.432/0.311 ms
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$
```

```
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$ ping 192.168.20.10
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=63 time=1.04 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=63 time=1.38 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=63 time=1.28 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=63 time=1.16 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=63 time=1.17 ms
64 bytes from 192.168.20.10: icmp_seq=6 ttl=63 time=1.08 ms
64 bytes from 192.168.20.10: icmp_seq=7 ttl=63 time=1.24 ms
64 bytes from 192.168.20.10: icmp_seq=8 ttl=63 time=1.50 ms
64 bytes from 192.168.20.10: icmp_seq=9 ttl=63 time=1.15 ms
^C
--- 192.168.20.10 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8012ms
rtt min/avg/max/mdev = 1.035/1.219/1.501/0.139 ms
```

Ping Successful with Server-VM (192.168.20.10)

Server VM (192.168.20.10):

Ping Successful with Router-VM (10.0.2.15)

```
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=6.03 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.626 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=1.59 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.686 ms
64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=0.689 ms
64 bytes from 10.0.2.15: icmp_seq=6 ttl=64 time=0.668 ms
^C
--- 10.0.2.15 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5098ms
rtt min/avg/max/mdev = 0.626/1.714/6.034/1.960 ms
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$
```

```
naimul-mannan@naimul-mannan-VirtualBox:~/Desktop$ ping 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=63 time=1.25 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=63 time=1.34 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=63 time=1.37 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=63 time=1.25 ms
64 bytes from 192.168.10.10: icmp_seq=5 ttl=63 time=1.28 ms
64 bytes from 192.168.10.10: icmp_seq=6 ttl=63 time=1.32 ms
64 bytes from 192.168.10.10: icmp_seq=7 ttl=63 time=1.10 ms
^C
--- 192.168.10.10 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6836ms
rtt min/avg/max/mdev = 1.103/1.272/1.369/0.080 ms
```

Ping Successful with Office Workstation-VM (192.168.10.10)

Summary:

- Built a virtualized multi-subnet network using VirtualBox and Ubuntu Server 22.04.
- Configured static IP addressing, subnet masks, and routing across two network segments.
- Implemented IP forwarding on a virtual machine acting as a router, and tested end-to-end connectivity using ICMP ping tests between devices on two different subnets
- Produced network topology documentation using [draw.io](https://draw.io) to demonstrate network architecture and planning.

#### Applications:

This project was done to simulate real life scenarios in corporations where they have to implement network segmentations between employee's computers from the servers. Also reinforced our understanding of routing, implemented our knowledge of IP forwarding to create a virtual router. Utilized ICMP to perform ping tests to test for connectivity between the two network segments. Successful ping tests confirmed that the routing tables and IP forwarding from the router are working properly.

#### Limitations/Issues: